



SHAH NEUROVISION
SPORTS TRAINING

LEVEL 01 · NEURAL WARMUP SERIES

QUIET EYE NEURAL WARMUP

A premium pre-performance protocol engineered to calm your gaze, sharpen visual fixation, and unlock the focus that separates elite athletes under pressure. Backed by sports neuroscience. Built for athletes who refuse to peak by accident.

FORMAT

PRINT + DIGITAL

DURATION

6-9 MIN

LEVEL

01 / BEGINNER



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SPORTS VISION · NEURO-VISUAL
PERFORMANCE

ATHLETE EDITION

V.01

INSIDE THIS GUIDE

WHAT YOU'LL TRAIN TODAY.

A complete Level 1 sequence — built from the same eye-tracking and gaze-control science used to train elite golfers, shooters, and pitchers. Run it before every practice and every game.

- | | | |
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| 01 | The Quiet Eye Principle
Why your gaze decides your performance under pressure. | P. 03 |
| 02 | Brain & Eyes — Working as One
The parieto-frontal network that powers elite vision. | P. 04 |
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What separates the 972 ms shooter from the 357 ms shooter. | P. 05 |
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01

THE PRINCIPLE

YOUR EYES ARE YOUR FIRST WEAPON.

What separates elite athletes under pressure is not just skill — it is **visual control**. Sports eye-tracking research uncovered a phenomenon called the **Quiet Eye**: a final, stable gaze on a target right before you move.

In golf, elite players lock onto the hole longer before the putt. In basketball, great shooters fix on one precise point of the rim before release. In baseball, elite hitters gather visual information earlier because their eyes are *controlled*, not scattered.

When stress rises, the eyes get jumpy. Attention breaks down. Timing follows. **Quiet Eye training** is how athletes engineer composure into their nervous system — before the whistle ever blows.

"Quiet eyes,
fewer jumps,
higher accuracy."

THE EDGE, IN NUMBERS

972 ms

ELITE FREE-THROW QUIET EYE

Average final fixation time of elite NCAA free-throw shooters before release.

357 ms

SUB-ELITE QUIET EYE

Sub-elite shooters fixate the rim for less than half the time — and miss more.

2.5–3.0 s

ELITE GOLF PUTT QUIET EYE

Tour-level putters hold a stable gaze nearly 2× longer than amateurs.

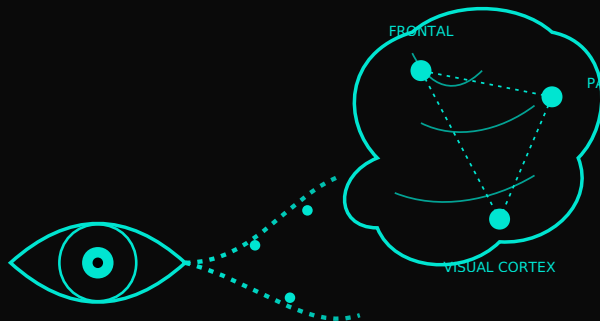
Source: Vickers (1996, 2007); Wilson & Pearcy; basketball QE meta-analyses.

02

THE PIPELINE

HOW YOUR BRAIN AND EYES WORK AS ONE.

5 LAYERS OF VISUAL PERFORMANCE



EYE / RETINA

LIGHT IN → PROCESSING → MOVEMENT OUT

Your eyes don't "see" — your brain does. The retina captures light, the optic nerve fires the signal back, and the **parieto-frontal network** turns that data into decisions and motor commands. Quiet Eye training tunes this whole pipeline.

L · 01

Visual Fixation

Holding your gaze on a single point without drift or jitter. The foundation everything else stacks on.

L · 02

Attention Control

Choosing what your brain processes — and what it ignores. Pressure attacks this layer first.

L · 03

Target Locking

The final, stable fixation on the exact point you're attacking — the literal "quiet eye."

L · 04

Eye Tracking

Smooth pursuit of a moving object — the ball, the opponent, the play developing.

L · 05

Neural Timing

The millisecond-level sync between what you see, when you decide, and when you move.

03

THE SCIENCE

WHAT THE RESEARCH ACTUALLY SHOWS.

In 1996, Dr. Joan Vickers tracked the eyes of elite and sub-elite basketball players. The result became the foundation of sports vision science: the **final fixation before motor action** — what she named the *Quiet Eye* — predicts performance more reliably than almost any other measurable variable.

Since then, the effect has been replicated in golf, shooting, archery, dart-throwing, surgery, and tactical decision-making. The pattern is consistent: **longer, steadier final fixations = higher accuracy under pressure.**

WHY IT WORKS

The brain needs runway.

A longer quiet eye gives your parieto-frontal network the milliseconds it needs to program movement parameters — force, direction, velocity — before your body executes. Less runway, sloppier landing.

FACT • 01

It's Predictive

QE duration predicts shot success even after controlling for skill and experience.

FACT • 02

It Trains Up

3–6 weeks of QE training reliably lengthens fixation and lifts accuracy in studies.

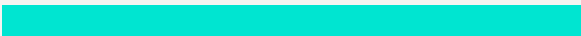
FACT • 03

It Survives Pressure

QE-trained athletes hold focus when pressure crashes other athletes' attention.

QUIET EYE DURATION • MS

ELITE FREE THROW

 972

SUB-ELITE FREE THROW

 357

ELITE GOLF PUTT

 2,500–3,000

AMATEUR GOLF PUTT

 1,000–1,500

0 750 1500 2250 3000

*Elite athletes hold a stable gaze 2–3× longer than sub-elite peers.
That fixation window is where accuracy is built.*

Quiet Eye Focus Technique

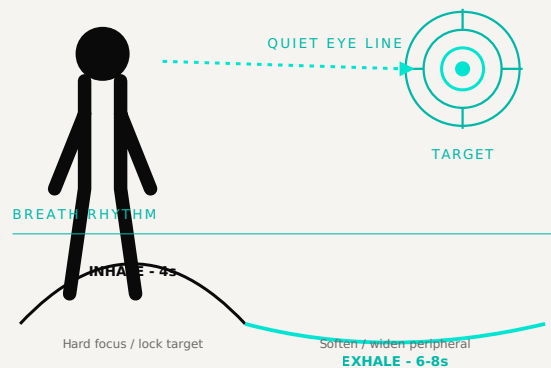
CYCLES 3-5 **TOTAL TIME** 90 S **EQUIPMENT** NONE **DIFFICULTY** ★ ★ ★ ★ ★

Purpose

Train your nervous system to settle the gaze on a single point and recover quickly when attention drifts. This is the master drill of the Level 1 series — every other drill stacks on it.

Step-by-Step

- 01 Pick a single, small target on the wall or in the distance — a screw head, a logo edge, a tree branch.
- 02 **Inhale 4 seconds** — hard-focus on every detail of the target. Read the texture, the edge, the shadow.
- 03 **Exhale 6-8 seconds** — widen your peripheral vision, let the target soften, feel your eyes go quiet.
- 04 At the bottom of the exhale, pause briefly. Notice the stillness.
- 05 Repeat for 3-5 full cycles. Build to 8 cycles as your tolerance grows.



COMMON MISTAKES

- Holding the breath instead of slow exhaling.
- Squinting — increases tension, defeats the drill.
- Choosing a target that is too large to fixate sharply.

Coaching Cues

- **"Eyes quiet, body loose."** The eyes lead the system — the body follows.
- **"One point. No scanning."** If the gaze starts hopping, return it gently.
- **"Soft on the exhale."** The widening is what builds composure.

WHY THIS WORKS NEUROLOGICALLY

You are training the parieto-frontal brake.

The slow exhale activates your vagus nerve and downshifts sympathetic arousal — the same system that makes your eyes jumpy under stress. Pair that with a stable visual fixation, and you are directly rehearsing the neural state of an elite athlete in the moment before execution.

Target Lock Saccades

SETS 2 × 30 S **REST** 15 S **EQUIPMENT** 2 TARGETS **DIFFICULTY** ★ ★ ☆ ☆ ☆

Purpose

Train your eyes to *land cleanly* on a new target after a saccade — the rapid jump movement. Elite athletes do not just look fast; they **arrive stable**.

Setup

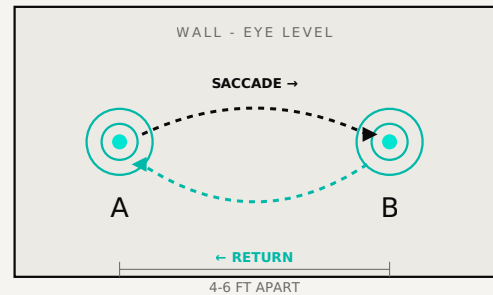
Place two small targets (sticky notes, coins, dots) about 4-6 feet apart on a wall, at eye level. Stand 6-8 feet back.

Step-by-Step

- 01 Lock onto Target A. Hold a quiet, stable fixation for one full second.
- 02 Snap your eyes to Target B as fast as possible — but the second you arrive, freeze the gaze.
- 03 Hold Target B silent for one second. No drift. No micro-jumps.
- 04 Snap back to Target A. Lock. Hold. Repeat.
- 05 Continue for 30 seconds, then rest 15 seconds. Do 2 sets.

Coaching Cues

- **"Snap, then silent."** Speed in. Stillness on arrival.
- **"Eyes only."** Keep your head perfectly still — isolate the eyes.
- **"No overshooting."** Land on the target, not past it.



TIMING PATTERN

LOCK 1s SNAP LOCK 1s SNAP LOCK 1s

Repeat for 30 seconds. Eyes only - head perfectly still.

COMMON MISTAKES

- Moving the head with the eyes — defeats the saccade isolation.
- "Sliding" the gaze instead of snapping. The jump must be ballistic.
- Overshooting the target, then drifting back — that is two errors stacked.

WHY THIS WORKS NEUROLOGICALLY

You are calibrating your superior colliculus.

Every saccade is computed in milliseconds by your midbrain's superior colliculus and frontal eye fields. Training the snap-then-silent pattern teaches your visual system to **arrive precisely** — instead of overshooting and re-aiming, the way an untrained gaze does under stress.

Near-Far Focus Shift

SETS

2 × 20

TEMPO

~1 / SEC

EQUIPMENT

PEN + WALL

DIFFICULTY

★ ★ ☆ ☆ ☆

Purpose

Train the eye's ability to shift focus between near and far targets cleanly — the same skill a hitter uses to pick up a pitch, a guard uses to read a defender, a quarterback uses to scan the field.

Setup

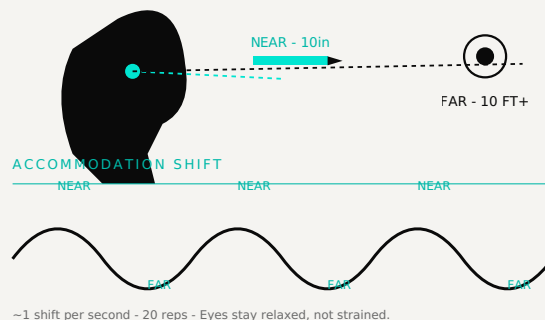
Hold a pen or finger about 10 inches from your nose. Choose a far target (10+ feet) directly behind it.

Step-by-Step

- 01 Focus sharply on the pen tip. Lock until the image is crisp — about 1 second.
- 02 Shift focus to the far target. Lock until *that* image is crisp.
- 03 Shift back to the pen. Crisp. Then back to far. Crisp.
- 04 Continue at ~1 shift per second. Complete 20 reps. Rest 20 seconds.
- 05 Do 2 sets. Each set, try to make the crisp-lock arrive faster.

Coaching Cues

- **"Crisp, not close enough."** Do not move until the image is sharp.
- **"Soft jaw, soft brow."** Tension shows up in the face first.
- **"Same speed both ways."** Most athletes are slower on near-to-far.



COMMON MISTAKES

- Shifting before the image is actually crisp.
- Tilting the head — keep it neutral, isolate the eyes.
- Going too fast — losing the lock kills the adaptation.

WHY THIS WORKS NEUROLOGICALLY

You are tuning the ciliary muscle & vergence system.

Every time you shift between near and far, your ciliary muscles change the shape of your lens and your eyes converge or diverge in millisecond unison. Trained athletes do this faster, cleaner, and with less fatigue — which is why they pick up the ball earlier and read plays a half-beat ahead of opponents.

Peripheral Awareness Anchor

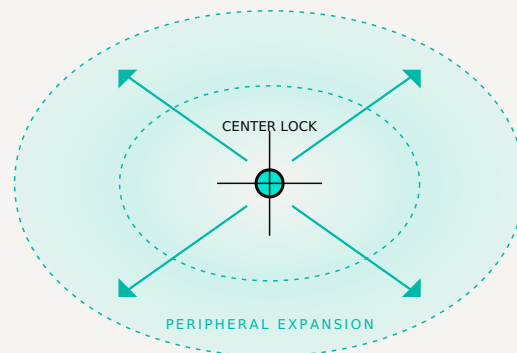
CYCLES 3 **HOLD** 30 S EACH **EQUIPMENT** NONE **DIFFICULTY** ★ ★ ☆ ☆ ☆

Purpose

Anchor your central fixation while consciously expanding peripheral awareness. This is the "calm wide" state — the visual signature of an athlete who is locked in but reading everything around the play.

Step-by-Step

- 01 Stand tall. Pick a single point straight ahead — eye level, ~10 feet away.
- 02 Lock central fixation. Without moving your eyes, expand your awareness *outward*.
- 03 Notice the room edges. Notice movement on the far left. Notice your hands at your sides — all without breaking the central lock.
- 04 Breathe slowly through the nose. 30 seconds total.
- 05 Rest 15 seconds, eyes closed. Repeat 3 times.



Eyes still. Awareness wide. 30 seconds.

COMMON MISTAKES

- Letting the eyes drift toward whatever you are peripherally noticing.
- Holding your breath while concentrating — kills the calm-wide state.
- Forcing the awareness — it should expand, not strain.

Coaching Cues

- **"Hard center, soft edges."**
- **"See without looking."** The eyes do not move. Awareness does.
- **"Wide is calm."** Narrow focus is fight-or-flight. Wide is composure.

WHY THIS WORKS NEUROLOGICALLY

You are shifting from narrow to ambient vision.

Under pressure, attention collapses into a narrow tunnel — a survival reflex. Peripheral training engages your **ambient visual system**, which is processed in parallel with your focal system. Athletes who train both stay composed, see plays develop earlier, and make decisions with more available information.

05

PRE-COMPETITION

MAKE IT YOUR PRE-GAME RITUAL.

Practice this sequence during training before you rely on it for games. Do not fumble through it cold.

Run it before every game and every training session — [consistency is what builds the neural pattern](#).

As you progress through the course, you will layer sport-specific drills on top of this base.

THE 90-SECOND SEQUENCE

STEP 01 -
0:00 - 0:20

Quiet Eye Focus

3 cycles - settle the nervous system, anchor the gaze.

STEP 02 -
0:20 - 0:45

Target Lock Saccades

1 set - prime the snap-and-stabilize pattern.

STEP 03 -
0:45 - 1:10

Near-Far Shift

1 set of 15 - prime the depth-shift system.

STEP 04 -
1:10 - 1:30

Peripheral Anchor

1 cycle - enter the calm-wide game state.

TIMELINE

0:00

0:20

0:45

1:10

1:30

QE FOCUS

TARGET LOCK

NEAR / FAR

PERIPHERAL

SAFETY CHECK

If anything feels off — stop.

If you experience pain, dizziness, persistent blur, double vision, or sharp headache during any drill — stop the protocol and check in with a vision specialist. This guide is a training tool, not a medical evaluation. Athletes recovering from concussion or with binocular vision conditions should consult Dr. Shah before running the program.

06

COMMON MISTAKES

THE ERRORS THAT CAP YOUR CEILING.

Most athletes do not fail at the moment of execution. They fail in the *120 milliseconds before it* — the window where the quiet eye should be doing its job. These are the patterns we see over and over.

MISTAKE - 01

Scanning Right Before Execution

Eyes hop from cue to cue right up to the moment of action. The brain never gets a stable fixation to compute the movement against.

Fix: Build a deliberate "final fixation" into your routine — one clean lock, then move.

MISTAKE - 02

Holding the Breath

Under pressure, athletes unconsciously hold breath. This jacks up sympathetic arousal and makes the gaze even jumpier.

Fix: Use the QE Focus exhale as your pre-execution cue. Long exhale = quiet eyes.

MISTAKE - 03

Looking at the Outcome, Not the Target

Shooting at "the rim" instead of one specific point on the rim. Throwing at "the strike zone" instead of one target.

Fix: Reduce the target. The smaller and more specific, the longer your quiet eye gets.

MISTAKE - 04

Skipping the Warmup on Game Day

Training the protocol all week, then "saving energy" and skipping it pre-game. The neural pattern needs to be primed every time.

Fix: Treat the 90-second protocol as non-negotiable. It is part of your gear.

MISTAKE - 05

Letting Pressure Narrow the Vision

Under stress, peripheral awareness collapses. Athletes "tunnel" and miss developing plays, open teammates, and incoming threats.

Fix: Use the Peripheral Anchor drill to rehearse the calm-wide state under increasing pressure.

MISTAKE - 06

Treating It Like Stretching, Not Skill

Going through the motions instead of training with intent. You cannot half-rep a neural protocol and expect a neural adaptation.

Fix: Every rep is a brain rep. Treat it like film study, not stretching.

07

QUICK REFERENCE

ONE PAGE.
TAPE IT TO YOUR BAG.

#	Drill	Sets / Time	Cue
01	Quiet Eye Focus 4s inhale - 6-8s exhale, single target	3-5 cycles - ~90s	<i>Eyes quiet, body loose.</i>
02	Target Lock Saccades 2 targets, 4-6 ft apart, eyes only	2 × 30s, 15s rest	<i>Snap, then silent.</i>
03	Near-Far Shift Pen at 10in, far target 10+ ft	2 × 20 reps	<i>Crisp, not close enough.</i>
04	Peripheral Anchor Central lock, expand awareness outward	3 × 30s, 15s rest	<i>Hard center, soft edges.</i>

THE MASTER CUE

"Quiet eyes = fewer jumps, more space, higher accuracy."

If you remember nothing else: *slow the gaze, lock the target, breathe out long.* That sequence — repeated, consistent, primed — is the single most reliable performance variable sports vision research has ever identified.

TOTAL TIME

~ 6 MIN

Full Level 1 protocol — all four drills, including rest. Use the 90-second express version on game day.

FREQUENCY

DAILY

Every training session. Every game. Neural adaptations come from repetition, not intensity.

08

PROGRESSION

HOW TO PROGRESS TO LEVEL 2.

You are ready to graduate when the Level 1 protocol feels automatic — when your gaze settles within a breath, when the saccades land clean every time, when the calm-wide state shows up without effort. That is usually **3-6 weeks** of daily practice. Do not rush it. Level 1 is the foundation everything else stacks on.

READINESS CHECKLIST

- You can hold a clean quiet eye for 3+ seconds without drift.
- Saccades arrive on-target without overshoot or re-aim.
- Near-far shifts feel effortless, not effortful.
- You can run the full 90s ritual without losing focus.
- You have done the daily protocol for at least 21 consecutive days.

LEVEL 2 ADDS

- Sport-specific quiet eye drills (rim, plate, hoop, goal).
- Dynamic gaze tracking against moving targets.
- Cognitive load drills - decision-making under fixation.
- Pressure inoculation - running the protocol under stress.
- Strobe and tachistoscopic flash work for processing speed.

FOR COACHES & PARENTS

This is not a quick fix. It is a system.

The athletes who get the most out of this protocol are the ones who treat it like skill work — same intent, same discipline, same consistency. Build it into the warmup. Stack it before every rep that matters. Over 3-6 weeks, you will see measurable changes in focus duration, reaction quality, and composure under pressure. That is not a placebo — that is neural adaptation.



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TRAIN - COMPETE - REPEAT

#QUIETEYE

What separates elite athletes under pressure is not just skill. It is **visual control**. Sports eye trackers revealed something called the "quiet eye" effect — where high-level athletes keep their gaze more stable before they move. That small difference changes timing, focus, and accuracy.

In golf, elite players lock onto the target longer before the putt. In basketball, great shooters stay visually fixed on one precise point of the rim before release. In baseball, elite hitters gather visual information earlier because their eyes stay controlled instead of scattered.

When stress rises, the eyes get jumpy. Attention breaks down. Timing follows. **That is why quiet eye training matters** for athletes who want better performance under pressure.

We use quiet eye drills inside our neural warmups to train focus, reaction timing, and visual stability before competition even starts.

Comment "QUIET EYE" and I will send you a Level 1 drill to use before games.

Watch the latest episode of the **NeuroVision Edge Podcast** for more sports neuroscience strategies that help athletes perform faster, calmer, and more focused under pressure.

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**SEE FIRST.
WIN OFTEN.**